

Enhancing Data Efficiency with a Trustworthy Counterfactual Generative Model

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Abstract—Leveraging limited data to synthesize an additional training set is essential for robotic vision, particularly in dynamic environments where collecting large datasets is impractical. Traditional robotic vision systems rely on extensive training data for object recognition and scene understanding but struggle to generalize to real-world variations such as lighting conditions, occlusions, and sensor noise. This paper proposes Causal DiffuseVAE, a novel method integrating causal inference with high-fidelity image synthesis to generate counterfactual images. By combining the disentanglement properties of Variational Autoencoders (VAEs) with the generative capabilities of Diffusion Models (DMs), Causal DiffuseVAE produces realistic, interpretable simulations of variations like shadows and occlusions. This combination enables data-efficient generative modeling by learning from small subsets and synthesizing missing or unseen samples. Additionally, causal inference ensures that generated data follow real-world dependencies, making it robust and interpretable for deployment in unpredictable environments. Four baseline approaches are evaluated across six different datasets, demonstrating that Causal DiffuseVAE consistently outperforms the four baseline approaches. This research elaborates the theoretical foundation and practical applications of Causal DiffuseVAE with the following project URL: <https://github.com/yza2542/Causal-DiffuseVAE>.

Index Terms—Counterfactual Image Generation, Causal Inference, Variational Autoencoder, Diffusion Models

I. INTRODUCTION

EFFICIENTLY processing and learning from data is becoming ever more crucial for perception and decision-making in robotics and other industrial applications [1]. Conventional robotic vision systems heavily rely on extensive, task-specific datasets for training [2]. Therefore, their deployment in dynamic real-world environments is limited because collecting large volumes of labeled data is impractical or cost-prohibitive [3]. Recent advances in generative modeling, such

This work was supported in part by the Royal Academy of Engineering/Leverhulme Trust Research Fellowship under Grant LTRF-2425-21-151, in part by the EPSRC Innovation Fellowship under Grant EP/S001956/2, and in part by the Royal Society-Newton Advanced Fellowship under Grant NAF\R1\201213.

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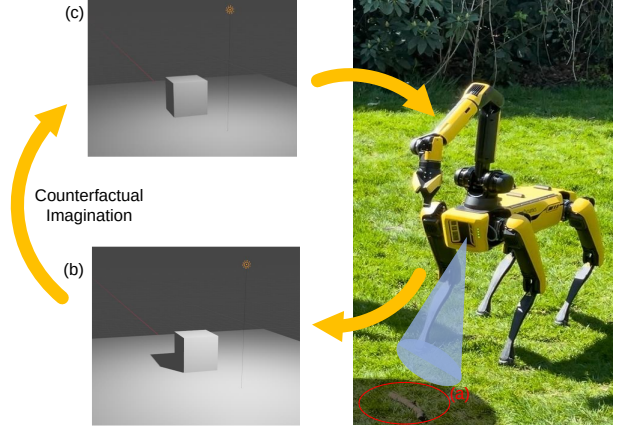


Fig. 1. A robotic dog is investigating a stick hidden in the shadow. (a) The stick under a shadow, (b) the scene's shadow geometry and light source, and (c) the counterfactual scene with the shadow removed.

as Variational Autoencoders (VAEs) and Diffusion Models, have made synthetic data generation more feasible. However, these models fall short when it comes to capturing and manipulating the causal factors underlying complex environmental variations [4]. Real-world images often suffer from degradations such as motion and blur in the defocus, which can further undermine perception and generative modeling [5]–[7]. How to obtain reliable data efficiently remains a challenge.

Causal inference, proposed to make the model interpretable, has gained popularity in machine learning in recent years [8]. Within causal inference, interventions and counterfactuals are deployed to examine how an event would develop under different conditions. An intervention explicitly alters one or more conditions of an event, while a counterfactual asks what would have happened under that intervention, given the actual observations [9]. Consider the example in Fig. 1, where a robotic dog must detect and grasp a stick hidden in shadow. A purely data-driven vision system may fail under such occlusions or lighting changes, leading to unsafe or incorrect decisions [10]. This highlights the need not only for high-quality generative models but also for causal counterfactual generation. Causal counterfactual generation enables asking “What if the shadow had been removed?” and then synthesizes the corresponding scene. Causal counterfactual generation provides targeted and realistic variations that improve model

interpretability under challenging conditions.

Causal inference in generative modeling has emerged as a promising solution to address interpretability and counterfactual reasoning. Structural Causal Models (SCMs) and related Directed Acyclic Graph (DAG)-based frameworks enable the explicit modeling of causal relationships among latent factors [11]. In recent works, VAEs are combined with masked causal layers that learn a latent adjacency matrix for disentangling exogenous factors, resulting in more interpretable embeddings, though at the cost of sample quality inherent to standard VAE decoders [12].

Diffusion-based Models, on the other hand, focus on photorealistic image synthesis by iteratively reversing a noise corruption process [13]. Although they have achieved high fidelity, general diffusion architectures lack explicit low-dimensional latents. They also typically require hundreds to thousands of denoising steps, making them computationally expensive at inference time and less amenable to direct causal manipulation [14]. Initial attempts to bridge causal inference and diffusion, such as Diff-SCM [15] and the Causal Diffusion Autoencoder (CDAE) [16], integrate causal constraints into the diffusion framework. Moreover, Diff-SCM operates directly in the high-dimensional diffusion space and uses gradient-based updates for both reasoning and intervention. However, Diff-SCM does not learn an explicit low-dimensional latent, making direct manipulation impossible. CDAE uses the Autoencoder (AE) to address the challenge that diffusion architectures lack explicit low-dimensional latents. AEs, which lack the Kullback–Leibler (KL) divergence regularization term found in VAEs, can “memorize” the training set and produce blurry or implausible reconstructions on novel inputs. In contrast, VAEs trade some reconstruction fidelity for a smoother, more robust latent embedding [17]. A more recent method, CausalFusion [18], integrates causal inference into Diffusion Transformers. However, the model only applies causal structure when generating samples and ignores any causal relationships between its internal representations. Furthermore, CCDiff introduces a compositional diffusion framework for closed-loop traffic scenario generation by embedding causal constraints into each denoising step [19]. Nevertheless, CCDiff assumes a fixed causal graph, which limits its ability to adapt to new or evolving causal structures.

This paper proposes the Causal Diffuse Variational Autoencoder (Causal DiffuseVAE), which integrates causal inference and high-fidelity image synthesis to enhance data efficiency by generating realistic and interpretable counterfactual images. The primary objective of this work is to enhance data efficiency for robots and autonomous vehicles by synthesising high-quality additional training samples in scenarios where real data are scarce. This model enables a vision system to learn from fewer real images by augmenting its training set with high-quality, causally controlled counterfactuals. The proposed method has exceptional capability in disentanglement and high-quality counterfactual image generation. These counterfactual images enable the vision system to imagine various scenarios, such as different shadow positions and intensities [20]. Furthermore, counterfactual images improve the overall efficiency of data utilization [21]. By generating trustworthy

counterfactuals that support efficient learning and adaptation, the proposed Causal DiffuseVAE provides both theoretical insight and practical feasibility for handling environmental variations such as shadows. The main contributions of this work are summarized as follows:

1) A structured latent space guided by an SCM is introduced, in which each latent dimension is set to correspond to a distinct causal factor. This method captures the causal relationships between factors, which helps producing counterfactuals that are semantically meaningful.

2) The interpretable latent encoding of a VAE is combined with the high-fidelity synthesis of a diffusion decoder, resulting in a single pipeline that benefits from both VAE and the Diffusion model under causal conditioning.

3) Identifiability of the causal latent is verified in the experiments. Evaluations on synthetic and real-world benchmarks show much stronger robustness against factor changes. The proposed method requires fewer annotated datasets.

The rest of this article is organized as follows. Section II introduces the relevant methods used in this paper. Section III describes the detail of the proposed model. Section IV represents the empirical results and theoretical analysis. Finally, Section V summarizes the article.

II. PRELIMINARIES AND PROBLEM FORMULATION

This paper considers the task of learning and generating counterfactual images by combining causal inference with high-fidelity diffusion decoding. In this framework, all the causal relationships are regarded as priors.

A. Intervention and Counterfactual in Structural Causal Models

SCM is commonly used in causal learning, and is defined with a triple $\langle U, V, E \rangle$ [22]. Here, U represents a set of exogenous variables determined by factors outside the model; V indicates the system’s internal variables; E represents the unexplained variation in an endogenous variable not captured by the model’s deterministic part. Model intervention is to convert an original image to a counterfactual image based on a causal graph through tuning one causal factor. Normally, causal inference is performed through interventions, typically represented by setting a variable to a fixed value regardless of its natural state. This is commonly expressed using the *do*-operator, denoted as $do(X = x)$, which implies actively setting X to x and observing the resultant changes in other variables [23]. The intervention aims to identify how changes in one variable (the cause) directly affect another variable (the effect). This process helps in answering counterfactual questions, such as “What would happen if we had changed X ?”

B. Mask Layer for Causal Inference

A mask causal layer is proposed to achieve causal disentanglement in [24]. The causal layer is to learn causal latent variables $\mathbf{z}$ with an adjacency matrix $\mathbf{A}$, which is obtained from a causal graph:

$$\mathbf{z} = \mathbf{A}^T \mathbf{z} + \epsilon = (\mathbf{I} - \mathbf{A}^T)^{-1} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (1)$$

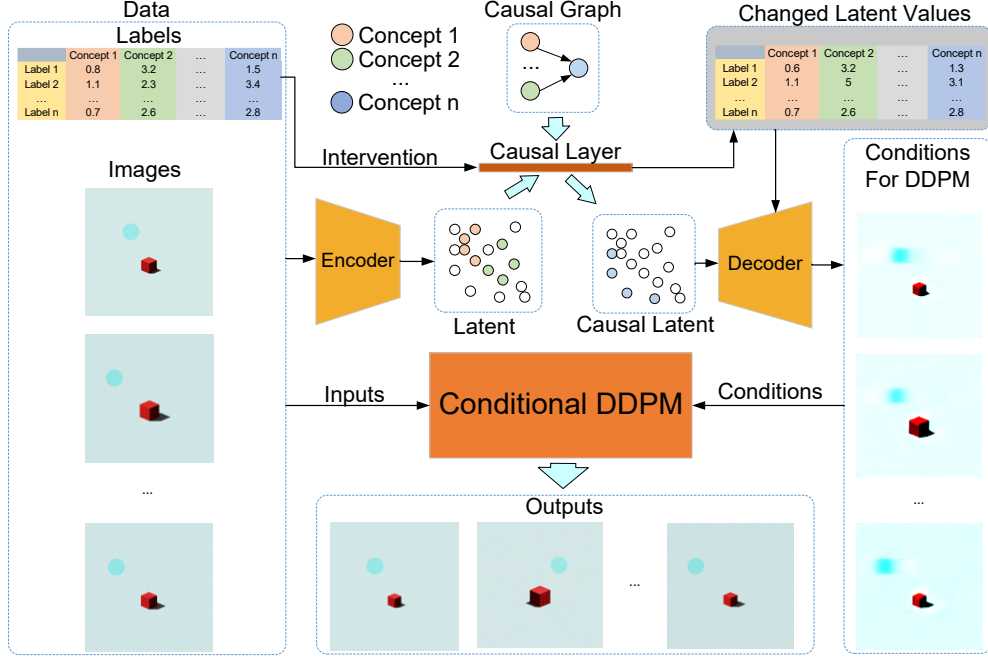


Fig. 2. Overall architecture of Causal DiffuseVAE. The labels and the images are the inputs of the model. Through the causal layer, the causal relationship among the latent can be learned in the training process. Furthermore, intervention can be performed by adjusting the values of the causal variables in order to influence the resulting labels of their effects.

where I is the identity matrix, ϵ indicates independent Gaussian exogenous factors and $\mathbb{T}$ denotes the transpose of a matrix. This masked causal layer enables the integration of causal inference with generative models.

C. Variational Autoencoders

VAEs are generative models that learn a generative mapping from a simple prior over a low-dimensional code to the data space, using an approximate inference network and the evidence lower bound (ELBO) objective [25]. A VAE defines a joint model over input images $\mathbf{x}$ and latent variables $\mathbf{z}$ [26]. The target distribution p is defined as

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I) \quad (2)$$

The conditional likelihood of the data given a latent variables $\mathbf{z}$ in the VAE is defined as

$$p_{\theta}(\mathbf{x} | \mathbf{z}) = f_{\theta}(\mathbf{z}) \quad (3)$$

where θ indicates the decoder parameters, and f_{θ} is a neural network that generates the parameters of the likelihood distribution for $\mathbf{x}$.

To approximate the intractable true posterior $p_{\theta}(\mathbf{z} | \mathbf{x})$, an encoder network is used to generate the parameters of a diagonal Gaussian from the approximate posterior q . The probability distribution $q_{\phi}(\mathbf{z} | \mathbf{x})$ of the VAE is defined as

$$q_{\phi}(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\phi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\phi}^2(\mathbf{x}))) \quad (4)$$

where ϕ indicates the encoder parameters and $\boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}_{\phi}(\mathbf{x})$ are neural networks mapping each input $\mathbf{x}$ to the mean and standard deviation of the approximate posterior. $\text{diag}(\boldsymbol{\sigma}_{\phi}^2(\mathbf{x}))$

is the covariance matrix of the Gaussian approximate posterior, with zero covariance off-diagonal.

ELBO is generally used as the loss function of the VAE

$$\begin{aligned} \mathcal{L}_{\text{VAE}} &= \text{ELBO} \\ &= \mathbb{E}_{q_{\phi}(\mathbf{z} | \mathbf{x})} [\log p_{\theta}(\mathbf{x} | \mathbf{z})] \\ &\quad - \text{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})) \end{aligned} \quad (5)$$

where $\mathbb{E}_Y[f(Y)]$ represents the function $f(Y)$ which is averaged over all possible values, weighted by the probability. The $\text{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z}))$ forces the approximate posterior toward the simple prior in (2). KL regularization encourages the model to use fewer latent dimensions. By avoiding a high KL penalty, the VAE sacrifices its ability to capture fine-grained image structure. This trade-off results in relatively low-quality generated images.

D. Diffusion Models

The original Diffusion Model, Denoising Diffusion Probabilistic Model (DDPM), was designed to model the process of transforming structured data into noisy data by adding Gaussian noise [27]. The probability distribution in the forward process of DDPM is defined as

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}) \quad (6)$$

where

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t I) \quad (7)$$

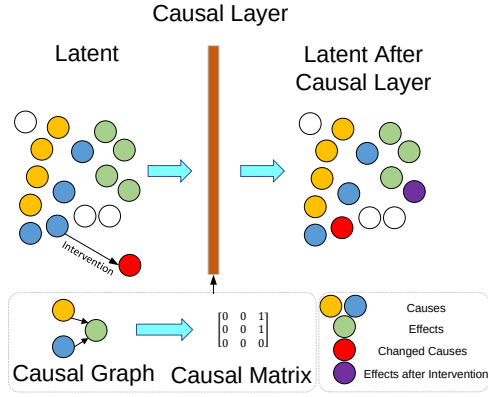


Fig. 3. Causal mechanism. The model has learned the causal relationship after training. When the cause is changed, the corresponding effect will also be changed.

while T represents the number of steps; x_0 donates the initial image; x_t represents the image at the step t ($t=1,2,\dots,T$); β_t is a predefined sequence of variance values that controls how much noise is added at each diffusion step. To reconstruct the structured data from noisy data, the target distribution in the reverse process of the DDPM is defined as

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p(\mathbf{x}_{t-1}|\mathbf{x}_t) \quad (8)$$

where

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\boldsymbol{\mu}(\mathbf{x}_t), \boldsymbol{\Sigma}(\mathbf{x}_t)) \quad (9)$$

while $\boldsymbol{\mu}(\mathbf{x}_t)$ and $\boldsymbol{\Sigma}(\mathbf{x}_t)$ are the mean and variance in Gaussian Distribution, respectively.

III. CAUSAL DIFFUSE VARIATIONAL AUTOENCODER

Current causal generative models, like CausalVAE [12] and CDRM [28], cannot generate 3-Dimensional images with both high accuracy and high quality due to the VAE's smooth loss function. The combination of a pixel-wise reconstruction term and the continuous KL divergence encourages averaged, smooth outputs and heavily penalizes sharp transitions. This loss function results in overly blurry reconstructions. The limitation leads to inconsistencies between generated images and the underlying engineering principles they aim to represent. The architecture of the Causal DiffuseVAE is illustrated in Fig. 2, which combines causal inference with the VAE and the Diffusion Model. Fig. 2 shows how data flow through an encoder, transforming inputs into causal latent representations. Causal mechanism specifies how one causal variable influences an effect variable, which is achieved by a causal layer. The causal layer organizes these latent variables based on an underlying causal graph, where labeled concepts, such as Concept 1, Concept 2, etc., are assigned numerical values. These values change when causal interventions are applied. Then the causal layer incorporates a decoder to generate conditions for the DDPM, which ensures the generative process of the DDPM respects causal dependencies. Furthermore,

model learning refers to the joint optimization of the encoder, causal layer, and diffusion network under the combined VAE and diffusion objectives. The causal mechanism and model learning are introduced below to show the details of the Causal DiffuseVAE.

A. Causal Mechanism

Causal Mechanism is to transform unstructured data into a structured latent space that explicitly captures and leverages the causal relationships among the underlying factors. This process operates under the mild assumption that all relevant causal variables are observed and that no confounders exist between them. In the proposed Causal DiffuseVAE, the causal latent variables $\mathbf{z}$ serve as the structured data, capturing the underlying causal factors that influence the observed features. A structured encoding process extracts $\mathbf{z}$, which maps labeled concepts to their respective causal influences. These structured latent variables then guide the Conditional DDPM, ensuring that generated images reflect the true causal dependencies. This approach enhances interpretability and reliability, allowing for meaningful interventions and counterfactual generation. In the Causal DiffuseVAE, $\mathbf{z}$ correspond to the causal features in the real world. Fig. 3 illustrates the mechanism of causal inference. During an intervention, when a specific latent in $\mathbf{z}$ is altered, the causal layer modifies the corresponding effect based on the encoded causal dependencies. The causal layer serves as a transformation matrix derived from a predefined causal graph. This causal layer encodes the relationships among $\mathbf{z}$, ensuring that changes in cause factors propagate their effects accordingly. Causal relationships are obtained from physical laws so the causal graphs are assumed to be fixed in the Causal DiffuseVAE. While the causal graph remains fixed, the causal matrix is designed to be trainable, allowing the model to adaptively refine its learned causal relationships throughout the training process. The mechanism of the causal inference is represented as

$$z_i = g_i(\mathbf{A}_i \circ \mathbf{z}; \eta_i) + \epsilon_i \quad (10)$$

where z_i is the i^{th} element of $\mathbf{z}$, g_i is the i^{th} element of the set of mild nonlinear and invertible functions $\mathbf{g}$, $\mathbf{A}_i$ is the i^{th} column vector in the adjacency matrix $\mathbf{A}$, $\circ$ is the elementwise product, and η_i is the learnable parameter of $\mathbf{g}$.

By enforcing $\mathbf{A}_i \circ \mathbf{z}$, we ensure that only the true parent nodes influence each z_i . This results in a disentangled, interpretable causal code as the true parent nodes capture the relationships between causes and effects. Because each z_i is built only from its masked parent nodes, any change to a "cause" latent will automatically propagate through all downstream nodes. This enforces that interventions on one latent spread through the network exactly following the causal arrows in the SCM.

Once $\mathbf{z}$ has been recomputed under these causal rules, it conditions the VAE decoder and downstream diffusion sampler. Because the decoder is explicitly trained to treat each dimension of $\mathbf{z}$ as a semantically meaningful factor, changes in a cause latent produce coherent and localized changes in the output image. In this way, the causal layer guarantees that

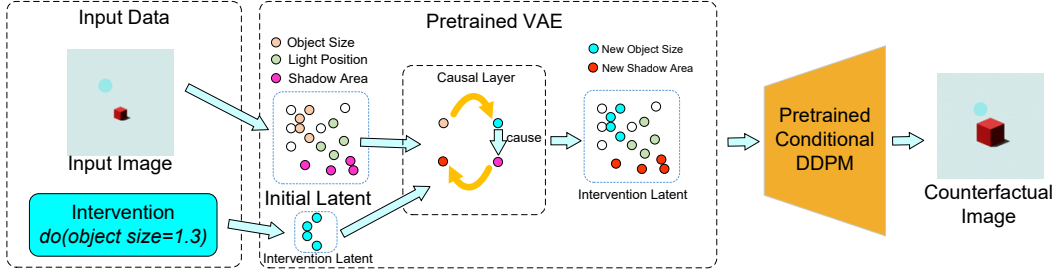


Fig. 4. The intervention process in the shadow situation. When the latent of the object size is changed, the shadow area in the causal layer will also change.

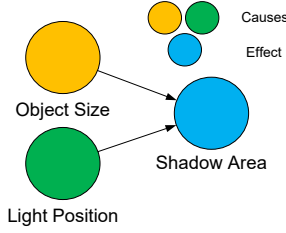


Fig. 5. The causal relationships in the shadow situation.

tuning a cause dimension leads to the correct effect in the generated image and reflects the learned causal graph.

Considering the case that a light source casts a shadow under an object as in Fig. 4, whose causal relationships are shown in Fig. 5. Let $\mathbf{z} = (z_{\text{light}}, z_{\text{object_size}}, z_{\text{shadow}})$, where z_{light} encodes the light position, $z_{\text{object_size}}$ indicates scale of the object, and z_{shadow} denotes the shadow area. If we increase $z_{\text{object_size}}$ as the intervention, the causal layer recomputes the shadow area

$$z_{\text{shadow}} = g_{\text{shadow}}(\mathbf{A}_{\text{shadow}} \circ \mathbf{z}) \quad (11)$$

As the object size increases, the corresponding shadow extends accordingly. The diffusion decoder generates an image in which the shadow shifts and enlarges, confirming that modifying the causal latent leads to the correct visual effect. These causal latents guide the diffusion decoder at every denoising step. By fusing this structured embedding with guidance information, the model follows the learned causal graph.

B. Model Learning

Model learning is to capture the complex causal dependencies within the observed data by refining both the structured latent space and the transformation functions. To model the evolution of latent variables across different contexts in the diffusion process, the loss function is defined as

$$L = \mathcal{L}_{\text{VAE}} + \mathcal{L}_{\text{DDPM}} \quad (12)$$

where $\mathcal{L}_{\text{DDPM}}$ is the loss function of the Diffusion Models.

1) *Learning Strategy of the VAE*: In $\mathcal{L}_{\text{VAE}}$, the generative process is defined as

$$p_{\theta}(\mathbf{x}, \mathbf{z}, \epsilon | \mathbf{u}) = p_{\theta}(\mathbf{x} | \mathbf{z}, \epsilon, \mathbf{u}) p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u}) \quad (13)$$

where $\mathbf{u}$ is the learning guidance. The inference process is defined as

$$q_{\phi}(\mathbf{z}, \epsilon | \mathbf{x}, \mathbf{u}) \equiv q(\mathbf{z} | \epsilon) q_{\zeta}(\epsilon - h(\mathbf{x}, \mathbf{u})) \quad (14)$$

where ζ is the vector of independent noise with probability densities q_{ζ} , and $h(\mathbf{x}, \mathbf{u})$ represents the mechanisms of the encoder.

In (5), the general ELBO is defined for uncausal latent. For the causal layer, the ELBO in the Causal DiffuseVAE is redefined to learn the parameters θ and ϕ as

$$\begin{aligned} \text{ELBO} = \mathbb{E}_{\mathbf{x}} \left\{ \underbrace{\mathbb{E}_{\epsilon, \mathbf{z} \sim q_{\phi}} [\log p_{\theta}(\mathbf{x} | \mathbf{z}, \epsilon, \mathbf{u})]}_{\text{Reconstruction Loss}} \right. \\ \left. - \underbrace{D_{\text{KL}}[q_{\phi}(\epsilon, \mathbf{z} | \mathbf{x}, \mathbf{u}) \| p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u})]}_{\text{KL Divergence Regularization}} \right\} \quad (15) \end{aligned}$$

The KL divergence between q_{ϕ} and p_{θ} is factorized as

$$\begin{aligned} D_{\text{KL}}[q_{\phi}(\epsilon, \mathbf{z} | \mathbf{x}, \mathbf{u}) \| p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u})] = \\ \mathbb{E}_{q_{\phi}} \left[\log \frac{q_{\phi}(\epsilon, \mathbf{z} | \mathbf{x}, \mathbf{u})}{p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u})} \right] \quad (16) \end{aligned}$$

The reconstruction loss is normally computed using Mean Square Error (MSE) or Binary Cross-Entropy (BCE). MSE produces high-quality images but smooths out causal latent structures by focusing on pixel-wise differences. BCE ensures high accuracy but reduces image quality as it treats pixel values independently. This mechanism makes the Causal DiffuseVAE better suited for binary data and potentially introduces artifacts in continuous image reconstruction. To generate images with both high quality and accuracy, we define the reconstruction loss $\mathcal{L}_{\text{recon}}$ in the Causal DiffuseVAE as

$$\mathcal{L}_{\text{recon}} = \alpha \cdot \text{BCE}(\mathbf{x}, \hat{\mathbf{x}}) + \nu \cdot \text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) \quad (17)$$

where $\hat{\mathbf{x}}$ is the reconstructed image; α and ν are the coefficients of the BCE and MSE, respectively, whose sum equals one.

Combined with (16) and (17), the reconstruction loss in the ELBO is replaced by using $\mathcal{L}_{\text{recon}}$, so the ELBO is derived as

$$\begin{aligned} \text{ELBO} = \mathcal{L}_{\text{recon}} - D_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u})) \\ = \alpha \text{BCE}(\mathbf{x}, \hat{\mathbf{x}}) + \nu \text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) - \mathbb{E}_{q_{\phi}} \left[\log \frac{q_{\phi}(\epsilon, \mathbf{z} | \mathbf{x}, \mathbf{u})}{p_{\theta}(\epsilon, \mathbf{z} | \mathbf{u})} \right] \quad (18) \end{aligned}$$

Besides, to ensure the identifiability of the adjacency matrix $\mathbf{A}$, the learning guidance $\mathbf{u}$ is used to guide the identifiability. $\mathbf{u}$ can be regulated by minimizing the loss function l_u :

$$l_u = \mathbb{E}_X \|\mathbf{u} - \sigma(\mathbf{A}^T \mathbf{u})\|_2^2 \quad (19)$$

where the notation $\|\cdot\|_2^2$ represents the squared L_2 -norm of a vector.

Learning with a similar method, $\mathbf{z}$ is regulated in the loss function by minimizing the loss function l_z :

$$l_z = \mathbb{E}_{\mathbf{z} \sim q_\phi} \left[\sum_{i=1}^n \|z_i - g_i(\mathbf{A}_i \circ \mathbf{z}; \eta_i)\|_2^2 \right] \quad (20)$$

To follow the training strategy to minimize the l_u and l_z , Causal DiffuseVAE minimizes the negating ELBO, which is unlike the standard VAE training objective of maximizing the ELBO. Overall, we have the new loss function

$$\mathcal{L}_{\text{VAE}} = -\text{ELBO} + \gamma l_u + \lambda l_z \quad (21)$$

where γ and λ are the regularization coefficients. Unlike a standard VAE which maximizes a true ELBO lower-bounding $\log p_\theta(x | u)$, our training loss directly minimizes the joint KL divergence in (16) plus causal regularizers. Consequently, the loss function of the VAE is not a strict bound on the marginal log-likelihood. However, because the objective remains a nonnegative divergence, gradient-based optimization still converges to a stationary point, and we empirically observe both stable training and improving sample quality throughout optimization. Incorporating the regularizers l_u and l_z in this framework allows us to enforce causal identifiability without sacrificing convergence stability. Therefore, we have

$$\begin{aligned} \mathcal{L}_{\text{VAE}} &= -\text{ELBO} + \gamma l_u + \lambda l_z \\ &= \underbrace{-\alpha \text{BCE}(\mathbf{x}, \hat{\mathbf{x}}) - \nu \text{MSE}(\mathbf{x}, \hat{\mathbf{x}})}_{\text{Reconstruction Loss}} \\ &\quad + \underbrace{\mathbb{E}_{q_\phi} \left[\log \frac{q_\phi(\epsilon, \mathbf{z} | \mathbf{x}, \mathbf{u})}{p_\theta(\epsilon, \mathbf{z} | \mathbf{u})} \right]}_{\text{KL Divergence Regularization}} \\ &\quad + \gamma l_u + \lambda l_z \end{aligned} \quad (22)$$

2) Learning Strategy of the Diffusion Model: In the Diffusion Model, $\mathbf{u}$ needs to be included. Specifically, the loss function will measure the discrepancy between the forward process and the reverse process. The forward process generates $\mathbf{x}_{1:T}$ conditioned on the additional information $\mathbf{y}$, $\mathbf{z}$, and $\mathbf{u}$, modeling how data evolve over time. The reverse process recovers $\mathbf{x}_0$ from $\mathbf{x}_{1:T}$ to reconstruct the original data. This discrepancy can be factorized as

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{\mathbf{z} \sim q_\psi(\mathbf{z} | \mathbf{y}, \mathbf{x}_0)} \left[\mathbb{E}_{q_\phi(\mathbf{x}_{1:T} | \mathbf{y}, \mathbf{z}, \mathbf{u}, \mathbf{x}_0)} \log \frac{q_\phi(\mathbf{x}_{1:T} | \mathbf{y}, \mathbf{z}, \mathbf{u}, \mathbf{x}_0)}{p_\psi(\mathbf{x}_{0:T} | \mathbf{y}, \mathbf{z}, \mathbf{u})} \right] \quad (23)$$

where ψ is the learnable parameter of the Diffusion Model. Therefore, the loss function in (12) is obtained by combining (22) and (23).

Algorithm 1 Causal DiffuseVAE Inference

Input: (image, label) pairs $(\mathbf{x}_0, \mathbf{u})$, number of the concepts n

Output: Generated counterfactual $\mathbf{x}_0^{\text{DM}}$

```

1: Sample  $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
2: for  $i = 1$  to  $n$  do
3:   if  $i$  = Intervention variable index then
4:      $\mathbf{z} = \text{Desired value}$ 
5:   else
6:      $\mathbf{z} = g_i(\mathbf{A}_i \circ \mathbf{z}; \eta_i) + \epsilon_i$ 
7:   end if
8: end for
9:  $\hat{\mathbf{x}}_0 \leftarrow p_\theta(\mathbf{x}_0 | \mathbf{z}, \epsilon, \mathbf{u})$ 
10: Add the condition,  $\mathbf{x}_T^{\text{DM}} = \mathbf{x}_T^{\text{DM}} + \hat{\mathbf{x}}_0$ 
11: for  $t = T$  to  $1$  do
12:   Sample noise  $\epsilon_t \sim \mathcal{N}(0, I)$ 
13:   Compute  $\mathbf{x}_t$  using:
14:    $\mathbf{x}_{t-1}^{\text{DM}} = (\mathbf{x}_t^{\text{DM}} - \sqrt{\beta_t} \cdot \epsilon_t) / \sqrt{1 - \beta_t}$ 
15: end for
16: Return  $\mathbf{x}_0^{\text{DM}}$ 

```

The Causal DiffuseVAE is a weakly supervised learning method. Although the learning guidance $\mathbf{u}$ is employed during training, the associated labels are not fully precise, as they do not indicate the exact spatial regions corresponding to the labeled features. This uncertainty in the labeling process contributes to the weakly supervised nature of the model, where the supervision is incomplete or inexact, making the learning process more challenging. With the pre-trained Causal DiffuseVAE model, the counterfactual images are generated by changing the value of one specific causal variable in n concepts. Then, the generating process continues, which is elaborated in Algorithm 1 with the following steps. First, sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$. For each concept index i , set z_i to the desired value if i is the intervention variable; otherwise compute (10). Next, combine $\mathbf{z}$, ϵ , and learning guidance $\mathbf{u}$ in the decoder to produce $\hat{\mathbf{x}}_0$. Afterwards, add $\hat{\mathbf{x}}_0$ to the noisy diffusion state $\mathbf{x}_T^{\text{DM}}$, and perform the reverse process for T steps by sampling ϵ_t and updating

$$\mathbf{x}_{t-1}^{\text{DM}} = \frac{\mathbf{x}_t^{\text{DM}} - \sqrt{\beta_t} \epsilon_t}{\sqrt{1 - \beta_t}} \quad (24)$$

The final output $\mathbf{x}_0^{\text{DM}}$ is the generated counterfactual image reflecting the applied causal intervention.

IV. EXPERIMENT

The experiments are deployed on a server with an Ubuntu 20.04 operating system and two NVIDIA RTX A6000 graphics cards. α and ν in (17) are set to 0.7 and 0.3, respectively. At the same time, γ and λ in (21) are set to 0.01 and 0.1, respectively.

A. Experimental Dataset

Six distinct datasets are selected for evaluation. In [29], a dataset generation method is proposed for shadow analysis.

Two synthetic shadow datasets were created using this method with Blender, each containing 10,000 images (7,500 for training, 1,000 for validation, and 1500 for testing). Each image includes a light source, an object, and its shadow. Variations in light source size and object properties systematically influence shadow formation, making the dataset ideal for causal analysis.

For general dataset evaluation, Causal DiffuseVAE was tested on the Pendulum and Flow datasets [12], each containing 7,000 images in Red, Green, Blue, and Alpha (RGBA) format. The Pendulum Dataset (5950 for training, 700 for validation, 350 for testing) captures causal interactions between pendulum angle, light angle, and shadow characteristics. The Flow Dataset (5,100 for training, 900 for validation, 1,000 for testing) simulates fluid dynamics as a ball interacts with liquid in a broken vessel, with transparency effects aiding visualization.

For real-world validation, the CelebA, a large scalable dataset, is used for facial attribute analysis [30]. The Gender and Age sub-dataset of the CelebA consists of 20,000 images, including 70% for training, 10% for validation, and 20% for testing. CelebA is widely used in face recognition, attribute prediction, and causal representation learning.

For validation in the industry situation, the Causal Circuit dataset was used [31]. The CausalCircuit dataset comprises 512×512 RGB images of a robot arm interacting with a causally connected circuit of buttons and lights, capturing four underlying causal variables. The dataset is divided into an 80% of training set and a 20% of test set, with no separate validation subset employed. This dataset demonstrates how models can model causal dependencies in a robotic control scenario, highlighting their potential applicability to real industrial perception and manipulation tasks.

B. Experimental setting

The most popular methods in the causal generative model are employed as baselines in evaluation, including CausalVAE [12], CDRM [28], CDAE [16] and Conditional Diffusion Models (CDM) [32]. The evaluation is based on two primary metrics: Mean Absolute Error (MAE) and Learned Perceptual Image Patch Similarity (LPIPS). MAE is employed to quantify the accuracy of the model’s control over latent factors:

$$\text{MAE} = (\sum_{i=1}^m |x_i - \hat{x}_i|) / m \quad (25)$$

where x_i is the original image, $\hat{x}_i$ is the generated image and m is the number of the tested images.

In evaluation, counterfactual images are generated and compared with Blender-rendered images using LPIPS, which measures perceptual similarity via feature representations rather than pixel differences. Lower LPIPS values indicate higher similarity. The LPIPS is defined as

$$\text{LPIPS}(x, \hat{x}) = \sum_l w_l \cdot \|F_l(x) - F_l(\hat{x})\|_2^2 \quad (26)$$

where $F_l(x)$ and $F_l(\hat{x})$ are feature representations extracted from a pre-trained network at layer l , and w_l are learned weights that adjust the contribution of each feature layer.

C. Experimental Results

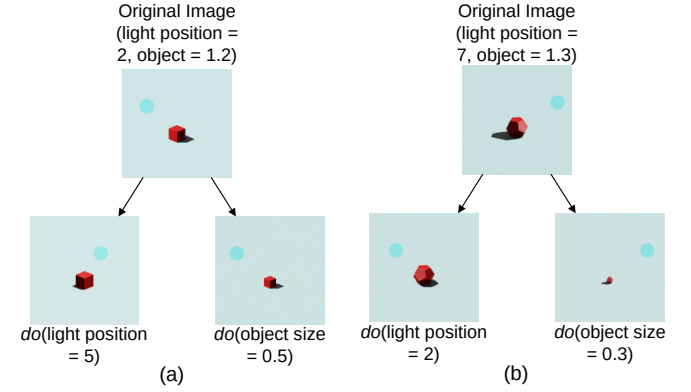


Fig. 6. Intervention results of shadow datasets. The object in (a) is the cube and in (b) is the polyhedron.

1) *Shadow*: Intervention results on the Shadow Dataset are presented in Fig. 6. Shadows cast by a cube and a polyhedron under light positions from 1 to 8 and object sizes from 0.3 to 1.7. As the light position and object size are varied, the shadow area is adjusted accordingly. The shape and the direction of the shadow areas demonstrate that Causal DiffuseVAE accurately models both shadow formation and object geometry.

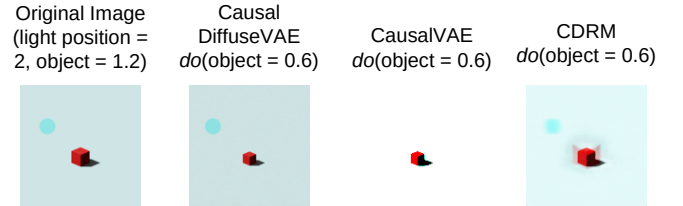


Fig. 7. Counterfactual images generated by Causal DiffuseVAE, CausalVAE and CDRM.

The intervention performance of the Causal DiffuseVAE, CausalVAE, and CDRM on the 3-dimensional dataset is presented in Fig. 7. When the object size changes, CausalVAE fails to generate the counterfactual 3-dimensional image due to the absence of the light source. CDRM successfully reconstructs all factors but fails to capture the underlying causal relationships. In contrast, Causal DiffuseVAE outperforms both methods by achieving high-quality reconstructions while accurately preserving causal dependencies.

2) *Flow*: Intervention results on the Flow Dataset are presented in Fig. 8, where ball sizes vary from 5 to 35, hole sizes from 0 to 4, water heights from 33 to 85, and water flow rates from 6 to 15. As the ball size is tuned, the water height is adjusted accordingly, which in turn influences the water flow. These variations follow physical laws, confirming that causal relationships have been accurately captured by Causal DiffuseVAE. Although CausalVAE and CDRM can also learn causal relationships, they fail to reconstruct the details, like the color and the hole of the vessel.

3) *Pendulum*: In the Pendulum Dataset, for each causal variable, the value of the label is in a fixed range. For the pendulum angle, the label values are set from -44 to 44, which

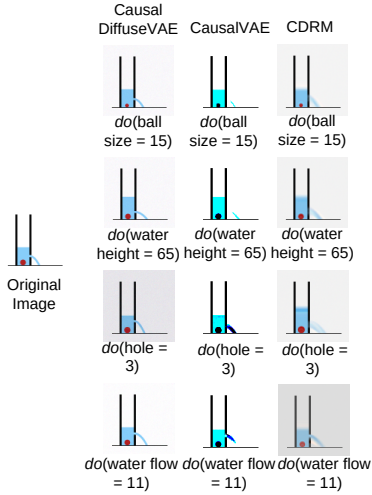


Fig. 8. Intervention results of flow datasets using Causal DiffuseVAE, CausalVAE and CDRM.

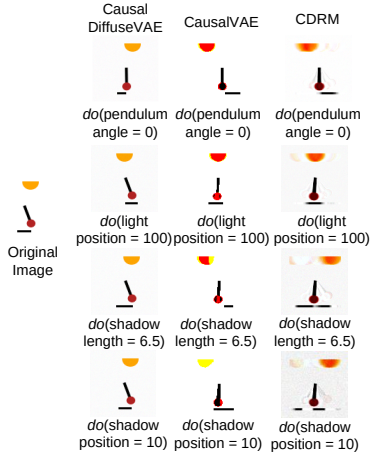


Fig. 9. Intervention results of pendulum datasets using Causal DiffuseVAE, CausalVAE and CDRM.

is the angle between the pendulum and the vertical line. For the light position, shadow length and shadow position, the label values are set from 60 to 140, 3 to 10, and 5 to 15, respectively. Fig. 9 illustrates the result of the intervention using Causal DiffuseVAE, CausalVAE and CDRM. Interventions on the pendulum angle and the light position lead to changes in the shadow length and shadow position. The results are similar to the results of the flow dataset. The CausalVAE and the CDRM fail to learn the details of the color and the shadow well.

4) *CelebA*: Fig. 10 shows the generated counterfactual images with the Gender and Age Dataset of the CelebA. Although the value of the labels is limited between -1 and 1, the “DO” operation can use the values from -1 to 1. When intervening the gender from -1 to 1, the faces in the generated images changed from female to male, which corresponds to the labels. With the change of gender, the hair and the beard are also changed. The female’s hair is more than the male’s. The female has no beard while the male has. When intervening the age from -1 to 1, the faces changed from an old person to a young person. The old person’s hair is less than the young’s.

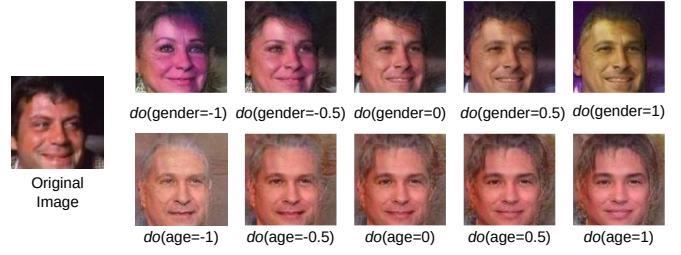


Fig. 10. Intervention results of CelebA datasets (Gender and Age) using Causal DiffuseVAE.

The beard on the old person’s face is clearer than the young person’s.

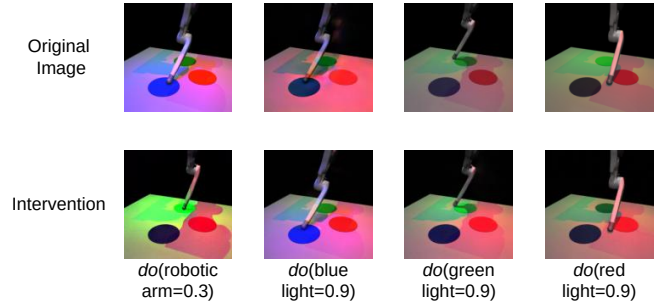


Fig. 11. Intervention results of Circuit datasets using Causal DiffuseVAE.

5) *Causal Circuit*: To assess the practical and industrial applicability of the proposed Causal DiffuseVAE, the model is further trained and evaluated on the Causal Circuit dataset, which simulates causal interactions in a robotic control environment. Fig. 11 shows the generated counterfactual images. Based on the causal relationships, when the robotic arm moves to a location, the corresponding light will be on or off. When the intensity of blue light or green light is changed, the intensity of the red light must be changed. When the intensity of the red light changed, the other lights would remain unchanged. As the intensity of the light increases, the light becomes brighter. In the results, when the robotic arm moves to the green light, the blue light is off, and the green light is on. When the intensity of blue light is tuned to 0.9, the red light and the blue light are on. The intervention of the green light is the same as the blue light. However, when the intensity of the red light is modified to 0.9, only the red light is on. These results follow the causal relationships among these factors.

6) *Result Analysis*: Table I shows that only Causal DiffuseVAE offers all of the following: an explicit low-dimensional causal latent; photorealistic outputs; stable end-to-end training via the reparameterization trick; reliable counterfactual generation even with incomplete observations; and a diffusion-based pipeline. No other baseline model combines this full set of capabilities. Table I uses objective rules to assign labels. A method supports an explicit low-dimensional causal latent if it exposes a dedicated latent vector that directly parameterizes causal factors. High-fidelity image synthesis is defined by whether blurring is observed in generated images.

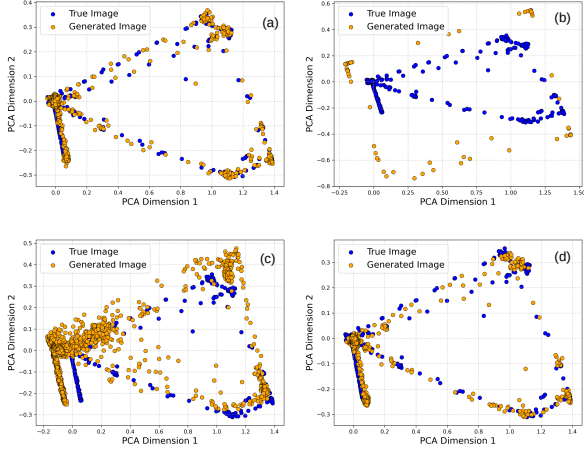


Fig. 12. Scatter plot of generated data and true data after PCA-based processing. (a) Causal DiffuseVAE. (b) CausalVAE. (c) CDRM. (d) CDAE.

Stable training via reparameterization requires the use of the reparameterization trick for unbiased gradient estimation. Trustworthy causal counterfactuals are supported when the model intervenes on a single latent factor and the generated image shows the expected change in that attribute. A diffusion-based architecture is characterized by a Markovian noise-adding forward process and a denoising reverse process as in DDPM.

By contrast, CausalVAE also provides an interpretable causal latent and benefits from stable, probabilistic training, but lacks the high-fidelity synthesis afforded by diffusion decoders. CDRM retains a structured latent space and supports stable training. CDRM is designed to impute missing entries and generate consistent counterfactuals from incomplete data. However, because CDRM relies solely on the VAE architecture, it cannot produce photorealistic images. CDAE recovers both a causal latent and high-quality diffusion synthesis, supporting reliable counterfactuals. CDAE lacks the VAE’s reparameterization-based encoder and full Bayesian uncertainty modeling. Finally conditional DDPMs excel at raw image fidelity. However, conditional DDPMs lack an explicit causal latent, cannot handle missing data in a principled way for counterfactual inference, and do not support stable, one-shot latent encoding.

At a high level, all of these models optimize composite loss functions that balance pixel-level fidelity with latent-space regularization. In the meantime, they also incorporate diffusion or imputation objectives to achieve both high-quality synthesis and principled causal reasoning.

Moreover, Causal DiffuseVAE jointly minimizes the revised ELBO of $\mathcal{L}_{VAE}$, in (21), and a latent-conditioned diffusion variational bound. An additional acyclicity penalty is comprised on its causal mask, producing reliable reconstructions and a well-structured DAG in the latent. CausalVAE and CDRM likewise optimizes a modified ELBO augmented by the same acyclicity constraint on its adjacency matrix, enforcing a valid causal graph over its latents but cannot generate high-quality images without diffusion-based model. CDAE adds a

supervised alignment loss on labeled semantic factors to its autoencoder, then conditions a Denoising Diffusion Implicit Model (DDIM)-style denoising objective on those factors to enable “DO” intervention counterfactual sampling. With the autoencoder, the model may generate overfitting results. Finally, conditional DDPMs are trained by minimizing the weighted diffusion variational bound across all timesteps, fitting a conditional denoiser for image synthesis but without any explicit causal or DAG regularization.

The principal component analysis (PCA)-based scatter plot analysis in Fig. 12 illustrates how effectively the model reconstructs and generates data while preserving causal structures. By reducing the dimensionality of both true and generated data, the visualization highlights the extent of alignment and potential deviations. The PCA result of the CausalVAE shows a relatively compact distribution, indicating strong latent representation but minor entanglement issues. Meanwhile, the PCA result of the CDRM maintains moderate alignment with true data but shows density shifts, suggesting challenges in precise causal modeling. At the same time, the PCA result of the CDAE demonstrates a structured yet slightly fragmented distribution, reflecting its denoising ability but potentially struggles in fully disentangling causal factors. For the result of the Causal DiffuseVAE, the generated data closely matches the true data distribution.

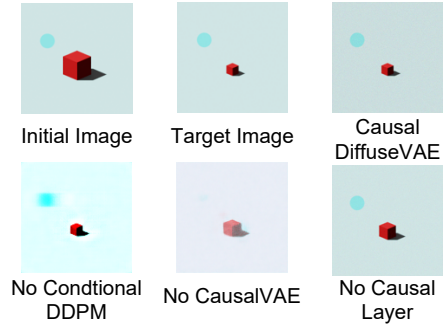


Fig. 13. Ablation results on the shadow dataset.

The results of the ablation experiments are summarized in Fig. 13. When the diffusion decoder is removed, blurring is observed in the images, indicating decreased quality. When both the VAE and causal layer are removed, color and lighting details are lost and object sizes are not adjusted correctly. When only the causal layer is removed, visual details are retained but adjustments of object size are unsuccessful. These findings confirm that each component of Causal DiffuseVAE is indispensable. The quantitative comparison in Table II shows that the full model achieves the best perceptual similarity, while each ablated setting leads to a degradation in image quality. These findings confirm that every component of Causal DiffuseVAE is essential for achieving high-fidelity and causally consistent generation.

As the result shown in Table III, Causal DiffuseVAE consistently outperforms other methods across all datasets, achieving the lowest error rates. This indicates that incorporating diffusion-based causal modeling enhances the model’s ability to capture structural dependencies and improve reconstruction

TABLE I. Comparison on capabilities of the Causal DiffuseVAE and other baseline methods

Capabilities	Causal DiffuseVAE	CausalVAE	CDRM	CDAE	conditional DDPM
Explicit low-dimensional causal latent	✓	✓	✓	✓	×
High-fidelity image synthesis	✓	×	×	✓	✓
Stable training via reparameterization	✓	✓	✓	×	×
Trustworthy causal counterfactuals	✓	✓	✓	✓	×
Diffusion-based architecture	✓	×	×	✓	×

TABLE II. LPIPS comparison of ablation results

Experiment settings	LPIPS
Causal DiffuseVAE	0.0185 ± 0.0140
Without Diffusion Decoder	0.0483 ± 0.0230
Without Causal Layer	0.0503 ± 0.0240
Without CausalVAE Module	0.1410 ± 0.1070

TABLE III. Comparison of the counterfactual images in evaluation using the MAE criterion

Dataset	Causal DiffuseVAE	CausalVAE	CDRM	CDAE	CDM
Cube Shadow	0.0098 ± 0.0050	0.0930 ± 0.0300	0.5400 ± 0.1300	0.2930 ± 0.0150	1.2100 ± 0.4590
Polyhedron Shadow	0.0103 ± 0.0040	0.1450 ± 0.0350	0.6600 ± 0.1320	0.4360 ± 0.0460	2.4500 ± 0.6770
Pendulum	0.0190 ± 0.0100	21.3020 ± 3.4400	17.9590 ± 2.5430	0.2980 ± 0.0100	0.8570 ± 0.3050
Flow	0.0230 ± 0.0100	16.6500 ± 3.3700	14.5900 ± 2.3220	0.3150 ± 0.0100	1.0100 ± 0.4680
CelebA (Gender)	0.0850 ± 0.0140	0.6500 ± 0.2400	0.7800 ± 0.4110	0.1340 ± 0.0369	0.9460 ± 0.3880

tion accuracy. In contrast, CausalVAE and CDRM exhibit significantly higher errors, particularly in complex datasets, highlighting their limitations in handling intricate variations. While CDAE and CDM show moderate performance, they still lag behind Causal DiffuseVAE, reinforcing the advantage of diffusion-based approaches in causal representation learning. These results demonstrate the potential of Causal DiffuseVAE in tasks requiring precise and robust generative modeling.

The results of the Causal DiffuseVAE and the baseline methods are listed in Table IV. Causal DiffuseVAE achieves the lowest perceptual similarity error in one dataset and remains highly competitive in the other. This indicates that the method preserves detailed structural information and produces perceptually similar reconstructions. CDAE also demonstrates impressive performance, particularly in one dataset, indicating its capability in certain scenarios. In contrast, CausalVAE, CDRM, and CDM exhibit higher perceptual errors, suggesting greater discrepancies between generated and original images. These findings reinforce the effectiveness of diffusion-based approaches in improving perceptual quality while maintaining causal consistency in image generation.

To evaluate the efficiency of the Causal DiffuseVAE and the baseline methods, the models are trained on 50% and 30% of the dataset and use LPIPS to assess the quality of the generated images. In the evaluation, the remaining data from the Shadow Dataset, unseen during training, are used for generation and comparison. These images are randomly selected from the unused portion of the dataset, ensuring that they are not included in the training process. Table V presents the LPIPS scores obtained under different training-data ratios. Causal DiffuseVAE consistently achieves the lowest LPIPS

TABLE IV. LPIPS comparison of counterfactual image quality

Dataset	Causal DiffuseVAE	CausalVAE	CDRM	CDAE	CDM
Cube Shadow	0.0185 ± 0.0140	0.1059 ± 0.0980	0.0590 ± 0.0400	0.0482 ± 0.0210	0.1410 ± 0.1070
Polyhedron Shadow	0.0292 ± 0.0160	0.1982 ± 0.1200	0.0680 ± 0.0400	0.0295 ± 0.0200	0.1976 ± 0.1300

TABLE V. LPIPS comparison of image quality under different training data ratios

Dataset	Causal DiffuseVAE	CausalVAE	CDRM	CDAE	CDM
Full dataset	0.0185 ± 0.0108	0.1059 ± 0.0980	0.0590 ± 0.0400	0.0482 ± 0.0217	0.1410 ± 0.1070
50% dataset	0.0193 ± 0.0113	–	0.0800 ± 0.0645	0.0735 ± 0.0483	0.1802 ± 0.1413
30% dataset	0.0235 ± 0.0150	–	0.1200 ± 0.0967	0.0917 ± 0.0657	0.2490 ± 0.1650

values, indicating stable generative quality even when trained with substantially reduced data. In contrast, CausalVAE fails to learn complete representations when trained on only 50% or 30% of the data. Moreover, other baselines exhibit much larger LPIPS increases as data decreases. These results demonstrate that Causal DiffuseVAE achieves high data efficiency by maintaining low LPIPS scores even when trained on just 50% or 30% of the data, whereas competing LPIPS of methods degrade sharply under the same data constraints.

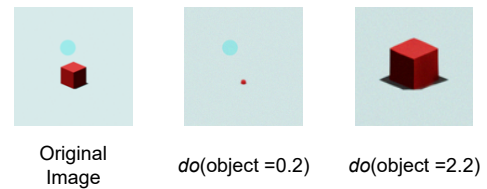


Fig. 14. Expanding intervention results of shadow using Causal DiffuseVAE.

Furthermore, the object size in the training data for the shadow dataset is limited from 0.3 to 1.7. To evaluate more on the efficiency of the Causal DiffuseVAE, the images are generated using the data out of the range, which is shown in Fig. 14. When the object fully hides the light source, Causal DiffuseVAE still predicts the correct shadow, demonstrating its generation capability outside the training range. The results in Fig. 14 and the Table V imply that Causal DiffuseVAE can effectively learn the causal data distribution from limited samples and generate the remaining data, thereby augmenting incomplete datasets. The extrapolation test in Fig. 14 confirms that the model can synthesize images beyond the labeled range, supporting its data-efficient generative capability.

In conventional diffusion models, high-resolution image generation is costly. Full pixel-space denoising is applied over hundreds or thousands of steps, causing significant compu-

TABLE VI. LPIPS scores of Causal DiffuseVAE, CDAE, and CDM at different sampling steps

Sampling Steps	Causal DiffuseVAE	CDAE	CDM
50	0.2156 $\pm$ 0.0580	0.3018 $\pm$ 0.0890	0.3566 $\pm$ 0.1930
100	0.0440 $\pm$ 0.0295	0.1486 $\pm$ 0.0470	0.2044 $\pm$ 0.1544
1000	0.0185 $\pm$ 0.0108	0.0482 $\pm$ 0.0217	0.1410 $\pm$ 0.1070

TABLE VII. Training and inference time of different models on the Shadow Dataset

Model	Training Time (h)	Inference Time (s/per image)
Causal DiffuseVAE	17.3	0.68
CausalVAE	2.0	0.66
CDRM	2.5	0.75
CDAE	55.2	1.8
CDM	17.4	6.6

tational and memory demands. By contrast, diffusion is performed within a compact latent space in Causal DiffuseVAE, reducing per-step complexity and memory footprint. As shown in Table VI, equivalent LPIPS scores are achieved in 50 steps for CDM instead of 1,000, and in 100 steps for CDAE instead of 1,000, corresponding to 20 $\times$ and 10 $\times$ fewer iterations. These findings demonstrate that high-resolution images can be generated with fewer iterations and higher data efficiency.

Table VII presents training and inference time measured on two RTX A6000 GPUs. Training and inference are performed quickly for CausalVAE and CDRM, but high-resolution images are not generated. Among models capable of high-resolution generation, training time is shorter for Causal DiffuseVAE and CDM than for CDAE, but inference time is longer for CDM and causal control cannot be performed. Inference time for diffusion-based models is defined as the time required to generate images that achieve the same LPIPS score. Under these matched-quality conditions, the inference of using Causal DiffuseVAE is faster, demonstrating higher data efficiency.

Across datasets, Causal DiffuseVAE achieves the lowest MAE and highest LPIPS, outperforming CDAE, CausalVAE, CDRM, and CDM by precisely controlling causal factors while preserving visual detail. These findings underscore the promise of diffusion-based causal modeling for realistic and interpretable image generation. Hence, the generated counterfactual images can provide more reliable scenes for vision systems. The various scenes improve scene understanding in dynamic environments, such as avoiding the disruptions of the shadow.

The proposed Causal DiffuseVAE demonstrates consistent advantages across multiple synthetic benchmarks, including the Causal Circuit dataset. However, a domain gap remains between these controlled settings and real-world deployment. Synthetic datasets enable precise control of causal variables such as light position, object geometry, and occlusion, which are difficult to isolate in physical experiments. However, real robotic perception must handle additional uncertainties, sensor noise, lighting variation, calibration errors, lens distortion, and dynamic environmental changes, which are not represented in synthetic scenes. Moreover, the causal inference framework employed in this work assumes that all relevant causal factors

are observed and that no unmeasured confounders influence the relationships among variables. This assumption is approximately satisfied in synthetic datasets where the data generation process is fully known, but may be violated in real-world conditions where latent environmental or hardware factors can act as hidden confounders. In the preparation of real-world datasets, the confounders may need to be identified and removed in advance to ensure that the observed variables maintain valid causal relationships.

To narrow the domain gap, several strategies can be employed. First, the pre-trained encoder-decoder can be fine-tuned on a small set of real sensor data using transfer learning to align latent causal factors with true sensory distributions. Second, domain randomization and photometric augmentation can be applied during synthetic data generation to approximate real-world variability in illumination and texture. Third, noise-robust or physics-aware causal encoders can be introduced to model measurement uncertainty explicitly within the structural causal graph. Fourth, hardware-aware optimization, such as weight quantization and pruning, can enable deployment on embedded robotic platforms without degrading fidelity. Future work will integrate these adaptation strategies and conduct validation on physical robotic systems, such as shadow-aware object grasping and light-conditioned manipulation. This will demonstrate that the counterfactual reasoning learned from synthetic environments can be transferred to real-world sensing and control tasks.

V. CONCLUSION

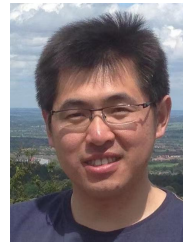
In this study, Causal DiffuseVAE, a novel generative method, is proposed that integrate causal reasoning into the VAE and Diffusion Model framework. Its effectiveness has been demonstrated through quantitative evaluations on the Cube Shadow and Polyhedron Shadow datasets. Experimental results indicate that Causal DiffuseVAE consistently outperforms conventional methods in both reconstruction accuracy and perceptual similarity. The method effectively preserves structural details while ensuring causal consistency. Under mild assumptions, identifiable and disentangled causal latents are produced by the model, ensuring consistent counterfactual generation. The reliability and interpretability of generated counterfactuals in real-world vision applications are supported by this rigorous theoretical foundation. Its capability to generate high-quality, causally controlled images shows its superiority in disentangling causal factors and reducing errors compared to CausalVAE, CDRM, CDAE, and CDM. The proven capability in generalization across different datasets suggests its potential for applications in shadow-aware vision systems and autonomous navigation, where accurate reconstruction and causal reasoning are essential. In future work, we will develop fully learnable causal graphs that update online with new data, allowing the model to adapt to evolving environments. We also plan to test Causal DiffuseVAE in real-world vision tasks like shadow removal, improving the robustness of perception in complex settings.

REFERENCES

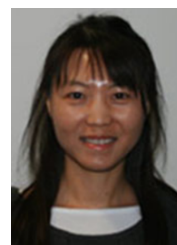
- [1] X. Li *et al.*, “Data Mode-Related Generative Adversarial Network for Industrial Soft Sensor Application,” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 3, pp. 4198–4205, 2024, doi: 10.1109/TII.2023.3319677.
- [2] S. Neupane *et al.*, “Security Considerations in AI-Robotics: A Survey of Current Methods, Challenges, and Opportunities,” *IEEE Access*, vol. 12, pp. 22072–22097, 2024.
- [3] Y.-T. Chen *et al.*, “On the Private Data Synthesis Through Deep Generative Models for Data Scarcity of Industrial Internet of Things,” *IEEE Transactions on Industrial Informatics*, vol. 19, no. 1, pp. 551–560, 2023, doi: 10.1109/TII.2021.3133625.
- [4] R. Hoque *et al.*, “IntervenGen: Interventional Data Generation for Robust and Data-Efficient Robot Imitation Learning,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 2840–2846.
- [5] K. Zhang *et al.*, “MC-Blur: A Comprehensive Benchmark for Image Deblurring,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 5, pp. 3755–3767, May 2024, doi:10.1109/TCSVT.2023.3319330.
- [6] K. Zhang *et al.*, “Adversarial Spatio-Temporal Learning for Video Deblurring,” *IEEE Transactions on Image Processing*, vol. 28, no. 1, pp. 291–301, Jan. 2019, doi:10.1109/TIP.2018.2867733.
- [7] K. Zhang *et al.*, “Deep Image Deblurring: A Survey,” *International Journal of Computer Vision*, vol. 130, no. 9, pp. 1–28, Sep. 2022, doi:10.1007/s11263-022-01633-5.
- [8] Y. Jung *et al.*, “Unified covariate adjustment for causal inference,” in *Advances in Neural Information Processing Systems*, vol. 37, pp. 6448–6499, Dec. 2024.
- [9] L. Cai *et al.*, “Counterfactual causal-effect intervention for interpretable medical visual question answering,” *IEEE Transactions on Medical Imaging*, vol. 43, no. 12, pp. 4430–4441, Dec. 2024, doi:10.1109/TMI.2024.3425533.
- [10] Z. Hau *et al.*, “Using 3D Shadows to Detect Object Hiding Attacks on Autonomous Vehicle Perception,” *2022 IEEE Security and Privacy Workshops (SPW)*, San Francisco, CA, USA, pp. 229–235, 2022, doi: 10.1109/SPW54247.2022.9833890.
- [11] A. Dhir *et al.*, “A Meta-Learning Approach to Bayesian Causal Discovery,” in *Proc. 13th Int. Conf. Learn. Represent. (ICLR)*, 2025. [Online]. Available: <https://openreview.net/forum?id=eeJz7eDWKO>
- [12] M. Yang *et al.*, “CausalVAE: Structured Causal Disentanglement in Variational Autoencoder,” *arXiv preprint*, Apr. 2020.
- [13] K. Sueyoshi and T. Matsubara, “Predicated diffusion: predicate logic-based attention guidance for text-to-image diffusion models,” in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2024, pp. 8651–8660, doi:10.1109/CVPR52733.2024.00826.
- [14] K. Preechakul *et al.*, “Diffusion Autoencoders: Toward a Meaningful and Decodable Representation,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022.
- [15] P. Sanchez and S. A. Tsaftaris, “Diffusion Causal Models for Counterfactual Estimation,” in *Proceedings of the 1st Conference on Causal Learning and Reasoning*, 2022.
- [16] A. Komanduri *et al.*, “Causal Diffusion Autoencoders: Toward Counterfactual Generation via Diffusion Probabilistic Models,” 2024. [Online].
- [17] H. Aetesam and S. K. Maji, “Deep variational magnetic resonance image denoising via network conditioning,” *Biomedical Signal Processing and Control*, vol. 95, p. 106452, 2024, doi:10.1016/j.bspc.2024.106452.
- [18] C. Deng *et al.*, “Causal Diffusion Transformers for Generative Modeling,” *arXiv preprint*, arXiv:2412.12095, Dec. 2024. [Online]. Available: <https://arxiv.org/abs/2412.12095>
- [19] Z. Lin *et al.*, “CCDiff: Causal Compositional Diffusion Model for Closed-loop Traffic Generation,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.
- [20] X. Tao *et al.*, “Counterfactual Reasoning and Cognitive Intelligence for Rational Robots,” in *2024 18th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, Dubai, United Arab Emirates, 2024, pp. 25–30.
- [21] P. Gupta *et al.*, “Object Importance Estimation Using Counterfactual Reasoning for Intelligent Driving,” *IEEE Robotics and Automation Letters*, vol. 9, no. 4, pp. 3648–3655, April 2024.
- [22] J. Dörfler *et al.*, “On the Complexity of Identification in Linear Structural Causal Models,” in **Advances in Neural Information Processing Systems** 37 (NeurIPS 2024), 2024, pp. 1–12.
- [23] G. Van Goffrier *et al.*, “Estimating long-term causal effects from short-term experiments and long-term observational data with unobserved confounding,” *arXiv preprint arXiv:2302.10625*, 2023.
- [24] J. Chung *et al.*, “Learning distribution-free anchored linear structural equation models in the presence of measurement error,” *Journal of the Korean Statistical Society*, vol. 54, pp. 361–385, 2025, doi:10.1007/s42952-024-00298-9.
- [25] S. Zhao *et al.*, “CV-VAE: A Compatible Video VAE for Latent Generative Video Models,” *arXiv preprint arXiv:2405.20279*, 2024.
- [26] S. Sadat *et al.*, “LiteVAE: Lightweight and efficient variational autoencoders for latent diffusion models,” in *Advances in Neural Information Processing Systems*, vol. 37, Dec. 2024.
- [27] J. Ho *et al.*, “Denoising Diffusion Probabilistic Models,” *arXiv preprint arXiv:2006.11239*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.11239>.
- [28] M. Chen *et al.*, “CDRM: Causal disentangled representation learning for missing data,” *Knowledge-Based Systems*, vol. 299, Article 112079, 2024.
- [29] J. Zhu *et al.*, “Shadow Datasets, New challenging datasets for Causal Representation Learning,” 2023. [Online].
- [30] Z. Liu *et al.*, “Deep Learning Face Attributes in the Wild,” in *Proceedings of the International Conference on Computer Vision (ICCV)*, Dec. 2015.
- [31] J. Brehmer *et al.*, “Weakly supervised causal representation learning,” in *Advances in Neural Information Processing Systems*, vol. 35, Dec. 2022.
- [32] P. Dhariwal and A. Nichol, “Diffusion Models Beat GANs on Image Synthesis,” in *Advances in Neural Information Processing Systems*, vol. 34, pp. 8780–8794, 2021. [Online].



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[Ye, Zhaoan](#), [Zhao, Dezong](#), Zhang, Li, [Yan, Xidong](#), [Bi, Qinglin](#), [Zhao, Wenjing](#) and [Flynn, David](#) (2026) Enhancing data efficiency with a trustworthy counterfactual generative model. *IEEE Transactions on Industrial Informatics*, (doi: [10.1109/TII.2026.3652484](https://doi.org/10.1109/TII.2026.3652484))



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Deposited on: 14 January 2026

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